

```

library(dplyr)

# Read the final analysis dataset
airline <- read.csv(
  "D:/data-portfolio/08_airline_service_performance_visual_analytics/data/processed/airline_performance.csv"
)

# Keep only network-level records
network <- airline %>%
  filter(Analysis_Level == "Network")

# Overall network performance
total_scheduled <- sum(network$Sectors_Scheduled)
total_flown <- sum(network$Sectors_Flown)
total_cancellations <- sum(network$Cancellations)

departure_otp <-
  sum(network$Departures_On_Time) / total_flown * 100

arrival_otp <-
  sum(network$Arrivals_On_Time) / total_flown * 100

cancellation_rate <-
  total_cancellations / total_scheduled * 100

cat("Total sectors scheduled:", total_scheduled, "\n")

## Total sectors scheduled: 7268896

cat("Total sectors flown:", total_flown, "\n")

## Total sectors flown: 7078663

cat("Weighted departure OTP:", round(departure_otp, 1), "%\n")

## Weighted departure OTP: 81 %

cat("Weighted arrival OTP:", round(arrival_otp, 1), "%\n")

## Weighted arrival OTP: 79.9 %

cat("Overall cancellation rate:", round(cancellation_rate, 2), "%\n")

## Overall cancellation rate: 2.62 %

# Annual network cancellation rates
annual_network <- network %>%
  group_by(Year) %>%
  summarise(

```

```

    Sectors_Scheduled = sum(Sectors_Scheduled),
    Cancellations = sum(Cancellations)
  ) %>%
  mutate(
    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100
  )

annual_network %>%
  filter(Year %in% c(2020, 2021))

```

```

## # A tibble: 2 x 4
##   Year Sectors_Scheduled Cancellations Cancellation_Rate
##   <int>         <int>         <int>         <dbl>
## 1  2020         228775         16400          7.17
## 2  2021         317252         33719         10.6

```

```

# Calculate monthly performance
monthly_network <- network %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100,

    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100,

    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100
  )

```

```

# Highest monthly cancellation rates
monthly_network %>%
  arrange(desc(Cancellation_Rate)) %>%
  select(
    Month,
    Cancellation_Rate,
    Sectors_Scheduled,
    Cancellations
  ) %>%
  head(5)

```

```

##           Month Cancellation_Rate Sectors_Scheduled Cancellations
## 1 2020-04-01      33.62306           6558           2205
## 2 2021-07-01      31.79423           29411           9351
## 3 2021-06-01      24.47376           38433           9406
## 4 2021-08-01      22.99172           16071           3695
## 5 2020-03-01      17.28375           43295           7483

```

```

# Lowest monthly departure OTP
monthly_network %>%
  arrange(Departure_OTP) %>%

```

```
select(
  Month,
  Departure_OTP,
  Arrival_OTP
) %>%
head(5)
```

```
##           Month Departure_OTP Arrival_OTP
## 1 2022-07-01      54.01178      54.96297
## 2 2022-06-01      61.90997      62.96816
## 3 2022-04-01      62.22673      63.59971
## 4 2022-05-01      64.17764      64.17764
## 5 2023-12-01      65.21378      63.60638
```

```
# Annual network performance and activity
annual_performance <- network %>%
  group_by(Year) %>%
  summarise(
    Sectors_Scheduled = sum(Sectors_Scheduled),
    Sectors_Flown = sum(Sectors_Flown),
    Departures_On_Time = sum(Departures_On_Time),
    Arrivals_On_Time = sum(Arrivals_On_Time),
    Cancellations = sum(Cancellations)
  ) %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100,

    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100,

    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100
  )
```

```
# Show annual performance from 2018 onwards
annual_performance %>%
  filter(Year >= 2018) %>%
  select(
    Year,
    Sectors_Scheduled,
    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    Cancellation_Rate
  )
```

```
## # A tibble: 7 x 6
##   Year Sectors_Scheduled Sectors_Flown Departure_OTP Arrival_OTP
##   <int>           <int>           <int>         <dbl>     <dbl>
## 1  2018             559095           548776         80.8       79.8
## 2  2019             566889           555233         79.9       78.4
## 3  2020             228775           212375         82.5       81.9
```

```
## 4 2021          317252          283533          85.9          86.2
## 5 2022          497284          473844          68.2          68.8
## 6 2023          543989          523593          71.0          70.6
## 7 2024           89389           86220          74.4          73.2
## # i 1 more variable: Cancellation_Rate <dbl>
```

```
# -----
# 4.2 Geographic airport performance
# -----

# Keep only route-level records
route_data <- airline %>%
  filter(Analysis_Level == "Route")

# Departure airport performance
departure_airport <- route_data %>%
  group_by(Departing_Port) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown),
    Departures_On_Time = sum(Departures_On_Time)
  ) %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100
  )

# Arrival airport performance
arrival_airport <- route_data %>%
  group_by(Arriving_Port) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown),
    Arrivals_On_Time = sum(Arrivals_On_Time)
  ) %>%
  mutate(
    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100
  )

# Highest-volume departure airports
departure_airport %>%
  arrange(desc(Sectors_Flown)) %>%
  select(
    Departing_Port,
    Sectors_Flown,
    Departure_OTP
  ) %>%
  head(10)
```

```
## # A tibble: 10 x 3
##   Departing_Port Sectors_Flown Departure_OTP
##   <chr>          <int>          <dbl>
```

```
## 1 Sydney          1248743      79.1
## 2 Melbourne       1024655      79.5
## 3 Brisbane        863020      82.1
## 4 Perth           359867      83.4
## 5 Adelaide        339227      84.1
## 6 Canberra        228645      83.4
## 7 Gold Coast      198138      75.8
## 8 Cairns          156833      81.9
## 9 Hobart          104347      74.0
## 10 Townsville     84898       83.1
```

```
# Share of sectors flown by the three largest departure airports
top_three_sectors <- departure_airport %>%
  arrange(desc(Sectors_Flown)) %>%
  head(3) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown)
  ) %>%
  pull(Sectors_Flown)

all_airport_sectors <- sum(departure_airport$Sectors_Flown)

top_three_share <-
  top_three_sectors / all_airport_sectors * 100

cat(
  "Share of sectors flown by top three departure airports:",
  round(top_three_share, 1),
  "%\n"
)
```

```
## Share of sectors flown by top three departure airports: 56.6 %
```

```
# Lowest departure OTP
departure_airport %>%
  arrange(Departure_OTP) %>%
  select(
    Departing_Port,
    Sectors_Flown,
    Departure_OTP
  ) %>%
  head(5)
```

```
## # A tibble: 5 x 3
##   Departing_Port Sectors_Flown Departure_OTP
##   <chr>          <int>          <dbl>
## 1 Sunshine Coast 47610          69.3
## 2 Ballina        24676          72.9
## 3 Hobart        104347          74.0
## 4 Port Macquarie 28144          75.0
## 5 Launceston    64173          75.1
```

```
# Highest departure OTP
departure_airport %>%
  arrange(desc(Departure_OTP)) %>%
  select(
    Departing_Port,
    Sectors_Flown,
    Departure_OTP
  ) %>%
  head(5)
```

```
## # A tibble: 5 x 3
##   Departing_Port Sectors_Flown Departure_OTP
##   <chr>          <int>          <dbl>
## 1 Port Lincoln      46643           89.4
## 2 Moranbah           8868           87.3
## 3 Bundaberg        12502           87.0
## 4 Newman           27484           86.0
## 5 Karratha          42138           85.2
```

```
# Lowest arrival OTP
arrival_airport %>%
  arrange(Arrival_OTP) %>%
  select(
    Arriving_Port,
    Sectors_Flown,
    Arrival_OTP
  ) %>%
  head(5)
```

```
## # A tibble: 5 x 3
##   Arriving_Port Sectors_Flown Arrival_OTP
##   <chr>          <int>          <dbl>
## 1 Port Macquarie  28221           74.4
## 2 Devonport       1559           75.8
## 3 Armidale        17281           76.7
## 4 Launceston      64271           76.9
## 5 Melbourne     1023554          77.0
```

```
# Highest arrival OTP
arrival_airport %>%
  arrange(desc(Arrival_OTP)) %>%
  select(
    Arriving_Port,
    Sectors_Flown,
    Arrival_OTP
  ) %>%
  head(5)
```

```
## # A tibble: 5 x 3
##   Arriving_Port Sectors_Flown Arrival_OTP
##   <chr>          <int>          <dbl>
## 1 Newman          27581           89.0
```

```
## 2 Port Lincoln          46702      88.9
## 3 Moranbah              9103       88.6
## 4 Port Hedland         30625       87.3
## 5 Bundaberg            12577       87.0
```

```
# Compare departure and arrival OTP for the same airport
```

```
airport_comparison <- departure_airport %>%
  rename(
    Airport = Departing_Port,
    Departure_Sectors_Flown = Sectors_Flown
  ) %>%
  select(
    Airport,
    Departure_Sectors_Flown,
    Departure_OTP
  ) %>%
  inner_join(
    arrival_airport %>%
      rename(
        Airport = Arriving_Port,
        Arrival_Sectors_Flown = Sectors_Flown
      ) %>%
      select(
        Airport,
        Arrival_Sectors_Flown,
        Arrival_OTP
      ),
    by = "Airport"
  ) %>%
  mutate(
    OTP_Gap =
      Arrival_OTP - Departure_OTP
  )
```

```
# Largest positive arrival minus departure OTP gaps
```

```
airport_comparison %>%
  arrange(desc(OTP_Gap)) %>%
  head(5)
```

```
## # A tibble: 5 x 6
##   Airport Departure_Sectors_Flown Departure_OTP Arrival_Sectors_Flown Arrival_OTP
##   <chr>          <int>          <dbl>          <int>          <dbl>
## 1 Sunshi~         47610          69.3          47771          77.5
## 2 Broome          23752          75.8          23730          83.7
## 3 Ballina         24676          72.9          24911          79.4
## 4 Hamilt~         21761          76.4          21801          82.8
## 5 Ayers ~          6545          75.3           6563          81.5
## # i abbreviated name: 1: Departure_Sectors_Flown
## # i 1 more variable: OTP_Gap <dbl>
```

```
# Largest negative arrival minus departure OTP gaps
```

```
airport_comparison %>%
```

```
arrange(OTP_Gap) %>%
head(5)
```

```
## # A tibble: 5 x 6
##   Airport Departure_Sectors_Flown Departure_OTP Arrival_Sectors_Flown Arrival_OTP
##   <chr>          <int>          <dbl>          <int>          <dbl>
## 1 Perth             359867             83.4             359649             79.9
## 2 Tamworth           9757             81.2              9810             78.1
## 3 Wagga Wagga       34535             82.4             34610             79.6
## 4 Mildura           31199             82.0             31248             79.5
## 5 Melbourne       1024655             79.5          1023554             77.0
## # i abbreviated name: 1: Departure_Sectors_Flown
## # i 1 more variable: OTP_Gap <dbl>
```

```
# -----
# 4.3 Airport performance across years
# -----
```

```
# Calculate annual arrival OTP for each airport
```

```
airport_year <- route_data %>%
  group_by(Arriving_Port, Year) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown),
    Arrivals_On_Time = sum(Arrivals_On_Time),
    .groups = "drop"
  ) %>%
  mutate(
    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100
  ) %>%
  filter(Year <= 2023)
```

```
# Network arrival OTP by year
```

```
network_year <- network %>%
  filter(Year <= 2023) %>%
  group_by(Year) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown),
    Arrivals_On_Time = sum(Arrivals_On_Time),
    .groups = "drop"
  ) %>%
  mutate(
    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100
  )
```

```
# Show network arrival OTP around the major decline
```

```
network_year %>%
  filter(Year %in% c(2021, 2022, 2023)) %>%
  select(
    Year,
```

```

    Arrival_OTP
  ) %>%
  mutate(
    Arrival_OTP = round(Arrival_OTP, 1)
  )

## # A tibble: 3 x 2
##   Year Arrival_OTP
##   <int>      <dbl>
## 1  2021         86.2
## 2  2022         68.8
## 3  2023         70.6

# Compare airport performance between 2021 and 2022
airport_2021 <- airport_year %>%
  filter(Year == 2021) %>%
  select(
    Arriving_Port,
    Arrival_OTP_2021 = Arrival_OTP
  )

airport_2022 <- airport_year %>%
  filter(Year == 2022) %>%
  select(
    Arriving_Port,
    Arrival_OTP_2022 = Arrival_OTP
  )

change_2021_2022 <- airport_2021 %>%
  inner_join(
    airport_2022,
    by = "Arriving_Port"
  ) %>%
  mutate(
    OTP_Change =
      Arrival_OTP_2022 - Arrival_OTP_2021
  )

# Number of airports declining from 2021 to 2022
cat(
  "Airports compared between 2021 and 2022:",
  nrow(change_2021_2022),
  "\n"
)

## Airports compared between 2021 and 2022: 36

cat(
  "Airports with lower arrival OTP in 2022:",
  sum(change_2021_2022$OTP_Change < 0),
  "\n"
)

```

```
## Airports with lower arrival OTP in 2022: 36
```

```
# Largest arrival OTP declines from 2021 to 2022
```

```
change_2021_2022 %>%  
  arrange(OTP_Change) %>%  
  head(5) %>%  
  mutate(  
    Arrival_OTP_2021 = round(Arrival_OTP_2021, 1),  
    Arrival_OTP_2022 = round(Arrival_OTP_2022, 1),  
    OTP_Change = round(OTP_Change, 1)  
  )
```

```
## # A tibble: 5 x 4
```

```
##   Arriving_Port Arrival_OTP_2021 Arrival_OTP_2022 OTP_Change  
##   <chr>          <dbl>          <dbl>          <dbl>  
## 1 Launceston      84.2            61.5          -22.7  
## 2 Melbourne       85.6            63.9          -21.6  
## 3 Rockhampton     87.3            65.8          -21.5  
## 4 Wagga Wagga     89.8            69.5          -20.3  
## 5 Gladstone       92              72.2          -19.8
```

```
# Compare airport performance between 2022 and 2023
```

```
airport_2023 <- airport_year %>%  
  filter(Year == 2023) %>%  
  select(  
    Arriving_Port,  
    Arrival_OTP_2023 = Arrival_OTP  
  )
```

```
change_2022_2023 <- airport_2022 %>%  
  inner_join(  
    airport_2023,  
    by = "Arriving_Port"  
  ) %>%  
  mutate(  
    OTP_Change =  
      Arrival_OTP_2023 - Arrival_OTP_2022  
  )
```

```
# Number of airports improving from 2022 to 2023
```

```
cat(  
  "Airports compared between 2022 and 2023:",  
  nrow(change_2022_2023),  
  "\n"  
)
```

```
## Airports compared between 2022 and 2023: 36
```

```
cat(  
  "Airports with higher arrival OTP in 2023:",  
  sum(change_2022_2023$OTP_Change > 0),  
  "\n"  
)
```

```
## Airports with higher arrival OTP in 2023: 29
```

```
# Largest recoveries from 2022 to 2023
```

```
change_2022_2023 %>%  
  arrange(desc(OTP_Change)) %>%  
  head(5) %>%  
  mutate(  
    Arrival_OTP_2022 = round(Arrival_OTP_2022, 1),  
    Arrival_OTP_2023 = round(Arrival_OTP_2023, 1),  
    OTP_Change = round(OTP_Change, 1)  
  )
```

```
## # A tibble: 5 x 4
```

```
##   Arriving_Port Arrival_OTP_2022 Arrival_OTP_2023 OTP_Change  
##   <chr>          <dbl>          <dbl>          <dbl>  
## 1 Proserpine    68            77.7           9.7  
## 2 Mount Isa     77.1          85.9           8.8  
## 3 Armidale      72.9          81.6           8.7  
## 4 Ballina       65.2          73.3           8.1  
## 5 Wagga Wagga   69.5          76.4           6.9
```

```
# Airports that declined while the network improved in 2023
```

```
change_2022_2023 %>%  
  filter(OTP_Change < 0) %>%  
  arrange(OTP_Change) %>%  
  mutate(  
    Arrival_OTP_2022 = round(Arrival_OTP_2022, 1),  
    Arrival_OTP_2023 = round(Arrival_OTP_2023, 1),  
    OTP_Change = round(OTP_Change, 1)  
  )
```

```
## # A tibble: 7 x 4
```

```
##   Arriving_Port Arrival_OTP_2022 Arrival_OTP_2023 OTP_Change  
##   <chr>          <dbl>          <dbl>          <dbl>  
## 1 Hamilton Island 77.6           71            -6.6  
## 2 Mackay          70.2           64.2          -6  
## 3 Alice Springs   78.8           75.2          -3.6  
## 4 Townsville      75.9           73.6          -2.3  
## 5 Emerald         83.5           81.4          -2.1  
## 6 Albury          76.4           75.4          -1  
## 7 Hobart          67.8           67.6          -0.2
```

```
# Keep airports with a complete 2010-2023 annual record
```

```
complete_airports <- airport_year %>%  
  group_by(Arriving_Port) %>%  
  filter(n_distinct(Year) == 14) %>%  
  ungroup()
```

```
# Count persistent high and low annual performance
```

```
airport_pattern <- complete_airports %>%  
  group_by(Arriving_Port) %>%  
  summarise(  
    
```

```

Years_Observed = n_distinct(Year),
Years_Above_85 = sum(Arrival_OTP >= 85),
Years_Below_80 = sum(Arrival_OTP < 80),
Mean_Annual_OTP = mean(Arrival_OTP),
.groups = "drop"
)

# Airports most frequently above 85% arrival OTP
airport_pattern %>%
  arrange(
    desc(Years_Above_85),
    desc(Mean_Annual_OTP)
  ) %>%
  head(5) %>%
  mutate(
    Mean_Annual_OTP = round(Mean_Annual_OTP, 1)
  )

```

```

## # A tibble: 5 x 5
##   Arriving_Port Years_Observed Years_Above_85 Years_Below_80 Mean_Annual_OTP
##   <chr>          <int>          <int>          <int>          <dbl>
## 1 Port Lincoln      14           12           1           88.8
## 2 Newman            14           11           0           88.9
## 3 Port Hedland      14           11           2           87.3
## 4 Kalgoorlie        14           10           3           86.1
## 5 Karratha          14           8            2           86.3

```

```

# Airports most frequently below 80% arrival OTP
airport_pattern %>%
  arrange(
    desc(Years_Below_80),
    Mean_Annual_OTP
  ) %>%
  head(5) %>%
  mutate(
    Mean_Annual_OTP = round(Mean_Annual_OTP, 1)
  )

```

```

## # A tibble: 5 x 5
##   Arriving_Port Years_Observed Years_Above_85 Years_Below_80 Mean_Annual_OTP
##   <chr>          <int>          <int>          <int>          <dbl>
## 1 Port Macquarie    14           1           13           74.9
## 2 Sydney            14           1           10           78.3
## 3 Launceston        14           0           9            77.2
## 4 Melbourne         14           1           9            77.3
## 5 Albury            14           1           9            78.2

```

```

# -----
# 4.4 Monthly performance in 2023
# -----

```

```

# Network monthly performance in 2023
network_2023 <- network %>%
  filter(Year == 2023) %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100,

    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100,

    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100
  ) %>%
  arrange(Month_Number)

# Show all network months in 2023
network_2023 %>%
  select(
    Month_Number,
    Month_Name,
    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    Cancellation_Rate
  ) %>%
  mutate(
    Departure_OTP = round(Departure_OTP, 1),
    Arrival_OTP = round(Arrival_OTP, 1),
    Cancellation_Rate = round(Cancellation_Rate, 2)
  )

```

##	Month_Number	Month_Name	Sectors_Flown	Departure_OTP	Arrival_OTP
## 1	1	Jan	42591	76.7	76.7
## 2	2	Feb	39220	73.7	73.0
## 3	3	Mar	44705	71.4	71.0
## 4	4	Apr	43421	71.2	71.8
## 5	5	May	44304	73.9	74.0
## 6	6	Jun	43163	70.3	69.0
## 7	7	Jul	44710	68.1	68.2
## 8	8	Aug	45148	73.2	72.5
## 9	9	Sep	44080	71.6	72.1
## 10	10	Oct	45205	71.1	71.1
## 11	11	Nov	43684	65.9	64.1
## 12	12	Dec	43362	65.2	63.6

##	Cancellation_Rate
## 1	3.14
## 2	3.85
## 3	3.57
## 4	3.94
## 5	3.01
## 6	3.63
## 7	3.96

```
## 8          3.82
## 9          3.62
## 10         3.77
## 11         3.69
## 12         4.96
```

```
# Calculate monthly arrival OTP for each airport in 2023
```

```
airport_month_2023 <- route_data %>%
  filter(Year == 2023) %>%
  group_by(
    Arriving_Port,
    Month_Number,
    Month_Name
  ) %>%
  summarise(
    Sectors_Flown = sum(Sectors_Flown),
    Arrivals_On_Time = sum(Arrivals_On_Time),
    .groups = "drop"
  ) %>%
  mutate(
    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100
  )
```

```
# Count airports below 70% arrival OTP in each month
```

```
airport_month_summary <- airport_month_2023 %>%
  group_by(
    Month_Number,
    Month_Name
  ) %>%
  summarise(
    Airports_Observed = n(),
    Airports_Below_70 =
      sum(Arrival_OTP < 70),
    Median_Arrival_OTP =
      median(Arrival_OTP),
    .groups = "drop"
  ) %>%
  arrange(Month_Number)
```

```
airport_month_summary %>%
  mutate(
    Median_Arrival_OTP =
      round(Median_Arrival_OTP, 1)
  )
```

```
## # A tibble: 12 x 5
##   Month_Number Month_Name Airports_Observed Airports_Below_70
##         <int> <chr>          <int>          <int>
## 1           1 Jan             36             3
## 2           2 Feb             36             9
## 3           3 Mar             36            12
```

```
## 4          4 Apr          36          11
## 5          5 May          37           5
## 6          6 Jun          37          12
## 7          7 Jul          37          18
## 8          8 Aug          37           5
## 9          9 Sep          37          10
## 10         10 Oct          37           7
## 11         11 Nov          36          26
## 12         12 Dec          36          21
## # i 1 more variable: Median_Arrival_OTP <dbl>
```

```
# Identify the ten highest-volume arrival airports in 2023
```

```
top_2023_airports <- airport_month_2023 %>%
  group_by(Arriving_Port) %>%
  summarise(
    Annual_Sectors_Flown =
      sum(Sectors_Flown),
    .groups = "drop"
  ) %>%
  arrange(desc(Annual_Sectors_Flown)) %>%
  head(10) %>%
  pull(Arriving_Port)
```

```
# Lowest monthly OTP observations among the ten busiest airports
```

```
airport_month_2023 %>%
  filter(
    Arriving_Port %in% top_2023_airports
  ) %>%
  arrange(Arrival_OTP) %>%
  select(
    Arriving_Port,
    Month_Name,
    Sectors_Flown,
    Arrival_OTP
  ) %>%
  head(10) %>%
  mutate(
    Arrival_OTP =
      round(Arrival_OTP, 1)
  )
```

```
## # A tibble: 10 x 4
```

```
##   Arriving_Port Month_Name Sectors_Flown Arrival_OTP
##   <chr>         <chr>         <int>         <dbl>
## 1 Brisbane     Dec           5282         56.3
## 2 Perth        Jul           2156          57
## 3 Melbourne    Nov           6705         57.7
## 4 Melbourne    Dec           6774         57.8
## 5 Perth        Dec           2258          60
## 6 Adelaide     Jun           2247          60
## 7 Townsville   Dec            455          60
## 8 Gold Coast    Dec           1459         60.5
## 9 Brisbane     Nov           5372         60.7
```

```
## 10 Sydney          Nov          7496          60.7
```

```
# -----  
# 4.5 Route and airline performance in 2023  
# -----  
  
# Calculate the 2023 network OTP benchmarks  
network_2023_benchmark <- network %>%  
  filter(Year == 2023) %>%  
  summarise(  
    Sectors_Flown = sum(Sectors_Flown),  
    Departures_On_Time = sum(Departures_On_Time),  
    Arrivals_On_Time = sum(Arrivals_On_Time)  
  ) %>%  
  mutate(  
    Departure_OTP =  
      Departures_On_Time / Sectors_Flown * 100,  
  
    Arrival_OTP =  
      Arrivals_On_Time / Sectors_Flown * 100  
  )  
  
departure_benchmark <-  
  network_2023_benchmark$Departure_OTP  
  
arrival_benchmark <-  
  network_2023_benchmark$Arrival_OTP  
  
cat(  
  "2023 network departure OTP benchmark:",  
  round(departure_benchmark, 1),  
  "%\n"  
)
```

```
## 2023 network departure OTP benchmark: 71 %
```

```
cat(  
  "2023 network arrival OTP benchmark:",  
  round(arrival_benchmark, 1),  
  "%\n"  
)
```

```
## 2023 network arrival OTP benchmark: 70.6 %
```

```
# Calculate route performance in 2023  
route_2023 <- route_data %>%  
  filter(Year == 2023) %>%  
  group_by(Route) %>%  
  summarise(  
    Sectors_Scheduled = sum(Sectors_Scheduled),  
    Sectors_Flown = sum(Sectors_Flown),  
    Cancellations = sum(Cancellations),  
    Departures_On_Time = sum(Departures_On_Time),
```

```

    Arrivals_On_Time = sum(Arrivals_On_Time),
    .groups = "drop"
  ) %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100,

    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100,

    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100,

    OTP_Gap =
      Arrival_OTP - Departure_OTP
  )

# Count routes relative to the 2023 network benchmarks
cat(
  "High departure and high arrival routes:",
  sum(
    route_2023$Departure_OTP >= departure_benchmark &
    route_2023$Arrival_OTP >= arrival_benchmark
  ),
  "\n"
)

```

High departure and high arrival routes: 45

```

cat(
  "Low departure and low arrival routes:",
  sum(
    route_2023$Departure_OTP < departure_benchmark &
    route_2023$Arrival_OTP < arrival_benchmark
  ),
  "\n"
)

```

Low departure and low arrival routes: 60

```

cat(
  "High departure and low arrival routes:",
  sum(
    route_2023$Departure_OTP >= departure_benchmark &
    route_2023$Arrival_OTP < arrival_benchmark
  ),
  "\n"
)

```

High departure and low arrival routes: 7

```
cat(
  "Low departure and high arrival routes:",
  sum(
    route_2023$Departure_OTP < departure_benchmark &
    route_2023$Arrival_OTP >= arrival_benchmark
  ),
  "\n"
)
```

```
## Low departure and high arrival routes: 14
```

```
# Highest-volume routes
route_2023 %>%
  arrange(desc(Sectors_Flown)) %>%
  select(
    Route,
    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    Cancellation_Rate
  ) %>%
  head(10) %>%
  mutate(
    Departure_OTP = round(Departure_OTP, 1),
    Arrival_OTP = round(Arrival_OTP, 1),
    Cancellation_Rate = round(Cancellation_Rate, 2)
  )
```

```
## # A tibble: 10 x 5
##   Route                Sectors_Flown Departure_OTP Arrival_OTP Cancellation_Rate
##   <chr>                <int>         <dbl>         <dbl>         <dbl>
## 1 Melbourne-Sydney    24222         72.5         67.6         8.95
## 2 Sydney-Melbourne    24159         68.6         66.7         8.84
## 3 Sydney-Brisbane     14887         69.9         68.5         5.65
## 4 Brisbane-Sydney     14864         73.7         71.4         5.82
## 5 Melbourne-Brisbane  11396         67.8         66.3         4.52
## 6 Brisbane-Melbourne  11339         68.4         67.2         4.68
## 7 Melbourne-Adelaide   8031         68.9         66.3         2.13
## 8 Sydney-Gold Coast    8004          69          68          3.72
## 9 Adelaide-Melbourne   8002         67.8         68.8         2.49
## 10 Gold Coast-Sydney   7997         68.3         67.8         3.99
```

```
# Highest cancellation rates
route_2023 %>%
  arrange(desc(Cancellation_Rate)) %>%
  select(
    Route,
    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    Cancellation_Rate
  ) %>%
```

```
head(10) %>%
mutate(
  Departure_OTP = round(Departure_OTP, 1),
  Arrival_OTP = round(Arrival_OTP, 1),
  Cancellation_Rate = round(Cancellation_Rate, 2)
)
```

```
## # A tibble: 10 x 5
##   Route                Sectors_Flown Departure_OTP Arrival_OTP Cancellation_Rate
##   <chr>                <int>         <dbl>         <dbl>         <dbl>
## 1 Melbourne-Sydney    24222         72.5         67.6         8.95
## 2 Sydney-Melbourne    24159         68.6         66.7         8.84
## 3 Canberra-Sydney     7418         74.5         73.1         8.17
## 4 Sydney-Canberra     7459         75.3         76.7         7.63
## 5 Armidale-Sydney     1501          80         76.9         6.42
## 6 Gladstone-Brisbane  2132         76.6         72.7         6.08
## 7 Sydney-Armidale     1495         75.3         81.6         5.97
## 8 Brisbane-Sydney    14864         73.7         71.4         5.82
## 9 Sydney-Brisbane     14887         69.9         68.5         5.65
## 10 Brisbane-Gladstone 2144         73.3         76.3         5.63
```

Largest positive OTP gaps

```
route_2023 %>%
  arrange(desc(OTP_Gap)) %>%
  select(
    Route,
    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    OTP_Gap
  ) %>%
  head(5) %>%
  mutate(
    Departure_OTP = round(Departure_OTP, 1),
    Arrival_OTP = round(Arrival_OTP, 1),
    OTP_Gap = round(OTP_Gap, 1)
  )
```

```
## # A tibble: 5 x 5
##   Route                Sectors_Flown Departure_OTP Arrival_OTP OTP_Gap
##   <chr>                <int>         <dbl>         <dbl>    <dbl>
## 1 Darwin-Adelaide      590         70.5         83.1     12.5
## 2 Brisbane-Perth      3458         63.2         73.4     10.2
## 3 Darwin-Brisbane     1072          62         71.5      9.5
## 4 Sydney-Darwin         62         56.5         64.5      8.1
## 5 Perth-Sydney        4660         60.8         68.6      7.8
```

Largest negative OTP gaps

```
route_2023 %>%
  arrange(OTP_Gap) %>%
  select(
    Route,
```

```

    Sectors_Flown,
    Departure_OTP,
    Arrival_OTP,
    OTP_Gap
  ) %>%
head(5) %>%
mutate(
  Departure_OTP = round(Departure_OTP, 1),
  Arrival_OTP = round(Arrival_OTP, 1),
  OTP_Gap = round(OTP_Gap, 1)
)

```

```

## # A tibble: 5 x 5
##   Route                Sectors_Flown Departure_OTP Arrival_OTP OTP_Gap
##   <chr>                <int>         <dbl>         <dbl>    <dbl>
## 1 Adelaide-Darwin      587           81.3           69     -12.3
## 2 Newman-Perth        1971           75.9           67.4     -8.5
## 3 Adelaide-Alice Springs 722           83.2           75.2      -8
## 4 Sunshine Coast-Sydney 2635           66.8           59     -7.7
## 5 Coffs Harbour-Sydney 2542           75.8           70.3     -5.5

```

```

# Calculate airline performance in 2023
airline_2023 <- airline %>%
  filter(
    Analysis_Level == "Airline",
    Year == 2023
  ) %>%
  group_by(Airline) %>%
  summarise(
    Sectors_Scheduled = sum(Sectors_Scheduled),
    Sectors_Flown = sum(Sectors_Flown),
    Cancellations = sum(Cancellations),
    Departures_On_Time = sum(Departures_On_Time),
    Arrivals_On_Time = sum(Arrivals_On_Time),
    .groups = "drop"
  ) %>%
  mutate(
    Departure_OTP =
      Departures_On_Time / Sectors_Flown * 100,

    Arrival_OTP =
      Arrivals_On_Time / Sectors_Flown * 100,

    Cancellation_Rate =
      Cancellations / Sectors_Scheduled * 100,

    OTP_Gap =
      Arrival_OTP - Departure_OTP
  )

# Compare all airlines
airline_2023 %>%

```

```

arrange(desc(Departure_OTP)) %>%
select(
  Airline,
  Sectors_Flown,
  Departure_OTP,
  Arrival_OTP,
  Cancellation_Rate,
  OTP_Gap
) %>%
mutate(
  Departure_OTP = round(Departure_OTP, 1),
  Arrival_OTP = round(Arrival_OTP, 1),
  Cancellation_Rate = round(Cancellation_Rate, 2),
  OTP_Gap = round(OTP_Gap, 1)
)

```

```

## # A tibble: 8 x 6
##   Airline      Sectors_Flown Departure_OTP Arrival_OTP Cancellation_Rate OTP_Gap
##   <chr>          <int>         <dbl>         <dbl>         <dbl>      <dbl>
## 1 QantasLink      138558          76.4          76.3           3.56      -0.1
## 2 Rex Airlines     71483          76.3          73.6           1.85      -2.7
## 3 Skytrans         2667          73.1          73.8           5.43       0.7
## 4 Qantas          91122          70.7          70.5           3.38      -0.3
## 5 Virgin Aust~  132506          67.2          66.2           4.21       -1
## 6 Virgin Aust~    5239          64.2          66.4           5.21       2.2
## 7 Jetstar        81318          64.1          65.8           5.06       1.7
## 8 Bonza          1094          62.6          64.2          12.7       1.6

```